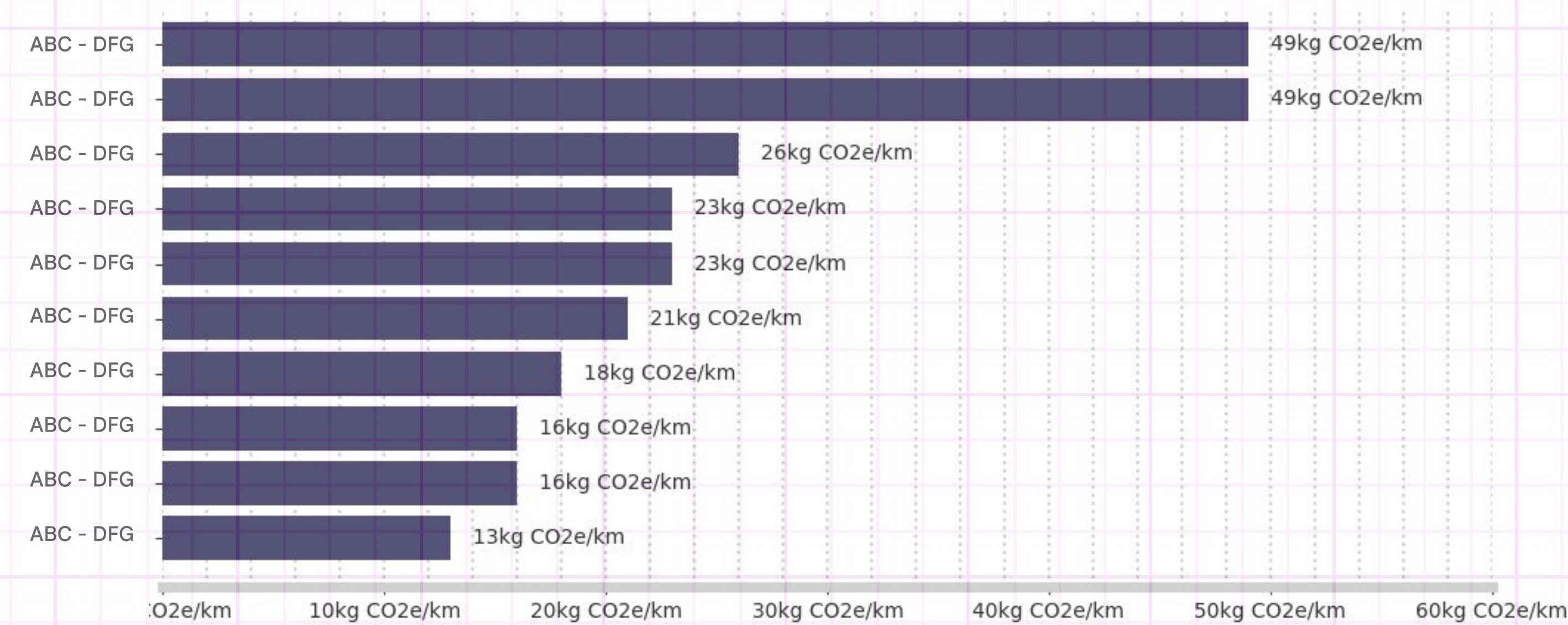


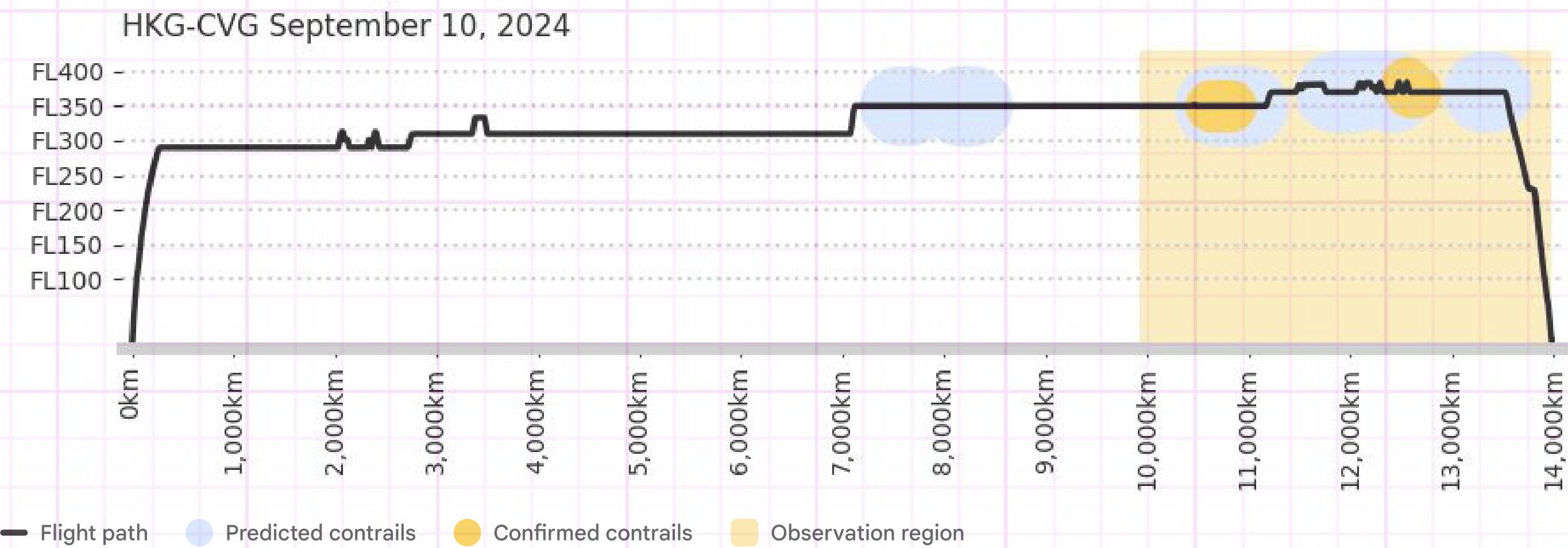
Origin-Destination pairs with the highest average total contrail warming per flown kilometer (CO2e/km) GWP50

The average carbon dioxide emissions per kilometer for [Airline] in September was 21 kg CO2/km. The OD pair with the highest contrail warming per kilometer is EMA - CPH adding 49 kg CO2e/km - or 2.3 times the average warming from the CO2 alone. The most warming OD pairs per flown kilometer are often flights that fly through contrail-prone zones (for example the Eastern part of the US) at night, when contrails are most warming.



Case study: predicted vs verified contrails

The light yellow color shows the observation area, and dark yellow indicates verified contrail formation from satellite imagery within the observation area. The light blue areas indicate where contrails are predicted to appear.



Did you know?

Some flight planning software providers, like [Flightkeys](#) and [CAE](#), have implemented contrail avoidance in their flight planning tools (or are about to).

In 2023, American Airlines, Google Research, and Reviate conducted a [trial](#) in which they avoided 54% of contrail kilometers by flying under contrail-prone areas.

In 2024, an [extensive study](#) of over 84,000 flights showed that, theoretically, it was possible to eliminate 73% of the contrail warming from these flights by spending 0.11% more jet fuel to adjust some of the flight paths.

See where contrails are forming right now on this [world map of contrails](#).

Read more about contrails on our websites: [Reviate](#), [Google Research](#).